

MPS-30 Series

Memory-Type Linear DC Power Supply

User Manual

MATRIX TECHNOLOGY INC.

Applicable models: MPS-31,MPS-32,
MPS-33,MPS-34,
MPS-35,MPS-36,
MPS-37,MPS-38,
MPS-39

Version: V1.1



Introduction

Dear User:

Hello! Thank you for purchasing the new MATRIX instrument. To ensure proper use of this device, please carefully read the entire instruction manual before operation, particularly the section on "Safety Precautions."

After reading the entire contents of this manual, we recommend storing it securely alongside the instrument or in a location where you can easily access it for future reference.

Copyright Information

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Verification and Correction Statement

The Company hereby expressly states that all instruments and equipment listed in this manual fully comply with the specifications and characteristics stated in the Company's technical documentation. These devices have undergone internal calibration prior to shipment, with the calibration procedures strictly adhering to the standards and protocols established by the Electronic Inspection Center.

Product quality is guaranteed.

Our company guarantees that all newly manufactured instruments undergo rigorous quality inspection. Within one year of shipment, we provide free repairs for any identified construction defects or component failures. However, free warranty service is not applicable if users have modified circuits or functions, repaired the instrument or its parts, or caused damage to the exterior packaging. Additionally, free warranty service is not provided for abnormal conditions resulting from improper grounding connections or failure to operate the equipment in compliance with safety regulations.

This warranty does not cover accessory equipment or other non-manufactured components for this instrument.

During the one-year warranty period, please return the faulty unit to our maintenance center or any authorized dealer; we will carry out proper repairs.

If this unit fails under abnormal operating conditions, due to human negligence, or caused by uncontrollable factors beyond human control (such as earthquakes, floods, riots, or fires), our company will not provide free warranty service.

(The Company adheres to a sustainable development strategy and reserves the right to improve the content of this manual without prior notice.)



Safety Precautions

During all stages of operating this instrument, the following general safety precautions must be observed. Failure to comply with these precautions or specific warnings outlined in other sections of this manual constitutes a violation of the safety standards pertaining to the instrument's design, manufacture, and intended use. Our company assumes no liability for users who fail to adhere to these precautions.

warn

- Do not use damaged equipment. Check that the equipment is in good condition before use. Do not operate this device in environments containing explosive gases, vapors, or dust.
- The instrument comes with a three-core power cord upon shipment, and its power supply unit must be connected to the three-core junction box. Before operating the power supply unit, ensure it is properly grounded to prevent accidental injuries!
- Before connecting the device, check all labels on it.
- Always use the cable provided with the device for connection to prevent accidental injuries.
- Use wires with appropriate rated loads; the capacity of all load wires must be sufficient to withstand the instrument's maximum short-circuit output current without causing overheating. If multiple loads are present, each pair of load wires must safely carry the instrument's full-rated short-circuit output current under overload conditions.
- To minimize fire and electric shock risks, ensure that voltage fluctuations in the mains power supply do not exceed 10% of the operating voltage range.
- When charging a battery with an instrument, always verify the correct polarity of the battery during wiring; otherwise, it may damage the device.
- Do not use this device if the cover is removed or loose.
- Do not install replacement parts on the instrument or make any unauthorized modifications.
- We shall not be liable for any direct or indirect financial losses that may arise from the use of this product.
- This equipment must not be used with life support systems or any other devices requiring strict safety precautions.
- Using the device in a manner not specified by the manufacturer may compromise the protection it provides.
- Always clean the device housing with a dry cloth. Do not clean the interior of the instrument.
- The instrument requires personnel supervision during operation. When not in use for an extended period, it should be turned off or disconnected from power supply.



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Chapter 1: Inspection and Installation

Before installation or operation, please review the safety signs and instructions in this manual.

1.1 Packaging List

Open the packaging and inspect the contents before operating the instrument. If any items are inconsistent, missing, or show signs of wear, contact the seller immediately.

Accessory Name	Quantity	Note Description
Programmable Instrument Host	1 PC	
Power line	1 PC	
Communication Line	1 PC	If the communication function is option- installed
Output cable	2 Pair	
User's manual	1 PC	
Product Factory Report	1 PC	



Note:

Confirm that the package contents are complete and intact. Keep the packaging box and relevant contents properly. The original packaging is required when the instrument is returned for factory service.

1.2 Installing the Instrument

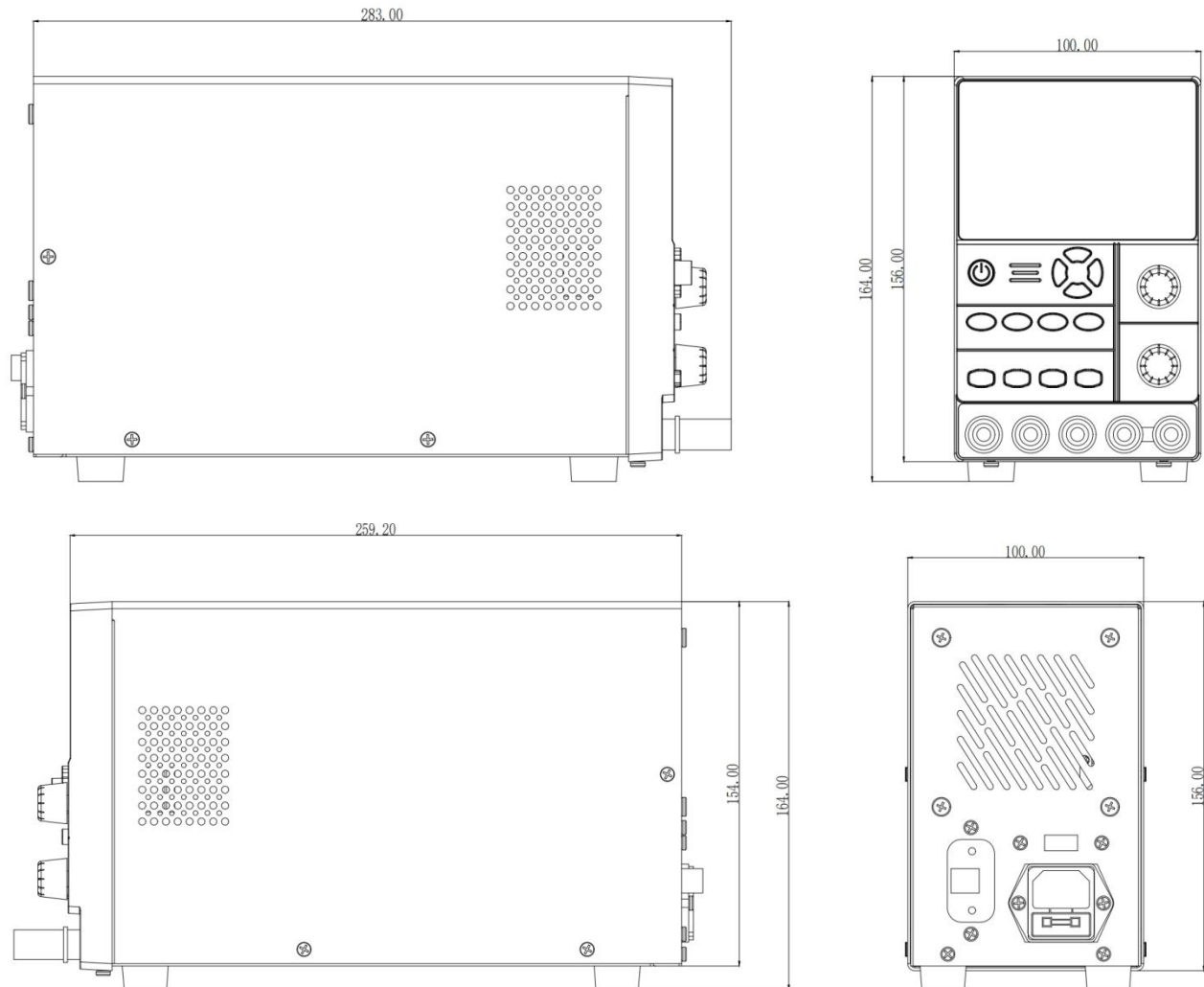
This instrument shall be installed in a well- ventilated space with adequate dimensions. Please select a suitable installation space according to the power supply dimensions specified below.

Dimensions of MPS-30 Series Programmable Power Supply:

100 mm (W)× 155 mm (H) × 259.2 mm (D) / 100 mm (W) × 155 mm (H) × 452 mm (D)

(W: Width; H: Height; D: Depth; H excludes feet, D excludes terminal posts)

Refer to the dimensions below for details.



Unit: Millimeter (mm)

1.3 Install the power cable

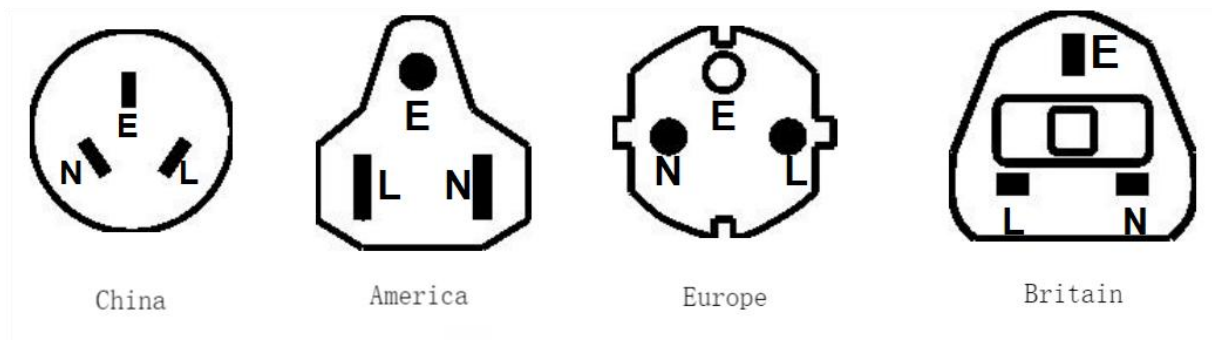
Connect the standard power cable to ensure the instrument is properly powered and reliably grounded.

Input Requirements for the Instrument

The power supply supports 110V and 220V operating voltages. Please check the input- voltage specification. A power- input cord matching your local mains is included in the accessories. If a mismatch is found, immediately contact the company's authorized dealer or after- sales service department.

Types of power cords

The power cord model provided by the power supply is shown in the figure below. Please select the power cord specification table below to choose the power cord suitable for the voltage of your region line model. If you make a mistake in the model when purchasing, please contact the dealer or directly find the manufacturer to replace it.



Chapter 2: Quick Introduction

This chapter briefly introduces the front- panel, rear- panel, keypad functions and TFT display functions of the MPS- 30 series power supply. It enables you to quickly get familiar with the appearance, structure and keypad operation before operating the power supply, so as to help you better operate this series of power supplies.

2.1 Introduction

The MPS-30 series programmable DC power supply is a programmable DC power supply with dual-channel / single- channel output. Featuring a clear user interface, excellent performance specifications and multiple optional communication interfaces, the MPS-30 series can meet diverse test requirements. Its key special functions and advantages are listed below:

- 3.5- inch high- visibility TFT display screen
- Adjust voltage and current separately via knob
- High response speed
- List output function
- Low ripple and low noise
- Intelligent fan control for energy saving and noise reduction
- CC/CV priority function
- Screen zoom- in display function
- Supports SCPI and MODBUS commands for easy construction of intelligent test platforms
- Over- voltage, over- current and over- temperature protection
- Fastest voltage rise time less than 10 ms

MPS-30 Series Programmable DC Power Supply Selection Table:

Model	Voltage	Current	Output Power
MPS-31	30V	2A	60W
MPS-32	30V	4A	120W
MPS-33	30V	6A	180W
MPS-34	30V	8A	240W
MPS-35	30V	10A	300W
MPS-36	60V	2A	120W
MPS-37	60V	4A	240W
MPS-38	80V	2A	160W
MPS-39	100V	0.5A	50W

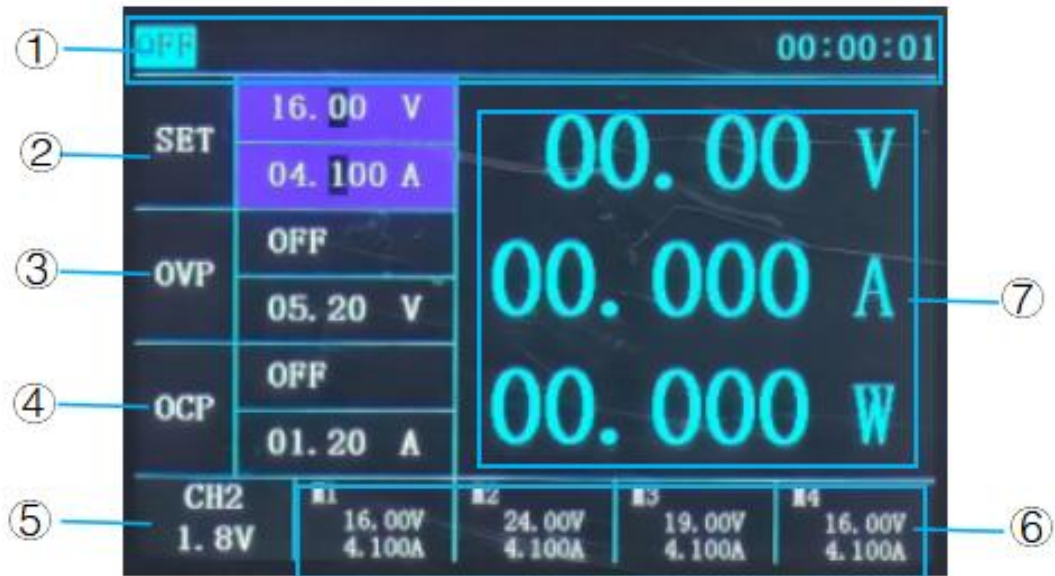
2.2 Introduction to the Front Panel

1. Introduction to the Front Panel



- ①. Instrument specification and model label
- ②. TFT display screen
- ③. Power switch
- ④. Function keys
- ⑤. Quick keys for storage and recall
- ⑥. CH1 output terminals (red for positive pole, black for negative pole)
- ⑦. Ground terminal
- ⑧. CH2 output terminals (red for positive pole, black for negative pole)
- ⑨. Current adjustment knob
- ⑩. Voltage adjustment knob (multi- function knob)

2.TFT Display Layout:



- ①.Power supply status status bar (displays output status, CV/CC/OVP/OCP/OTP/PC-connection/ runing time, etc.
- ② CH1 output voltage and current parameter setting (Refer to Section 3.2, 3.3 Voltage and Current Setting)
- ③ CH1 over- voltage setting (Refer to Section 3.5.1 Over- Voltage Setting)
- ④ CH1 over- current setting (Refer to Section 3.5.2 Over- Current Setting)
- ⑤ CH2 voltage setting parameter display window
- ⑥ Display window for voltage and current values stored in M1, M2, M3, M4
- ⑦ CH1 output voltage, current and power parameter display bar

2.3 Keypad Introduction



The key assignments are as follows:

Key	Name and Function
	Power on/off key
Menu	Menu function key for setting relevant power supply parameters
List	List key for configuring List- mode parameters
Lock	Keypad lock function key. Press once to lock panel keys; long- press again to unlock
Output	Output On/Off key, controls output enable/disable for CH1 and CH2
	Up, down, left, right direction keys; move cursor to target position under setting mode
	Multi- function knob for setting relevant power supply parameters
M1,M2, M3,M4	M1, M2, M3, M4 shortcut keys, used to store and recall frequently- used voltage and current parameters

2.4 Introduction to the Rear Panel

Instrument Rear Panel Layout:



- ① Heat- dissipating ventilation opening
- ② Power socket with fuse holder (AC power input interface)
- ③ Communication port (optional accessory)

2.5 Power-on Self-check

A successful self- test indicates that the instrument you purchased complies with factory specifications and is ready for normal operation. Before operating the instrument, please make sure you have read and understood the safety instructions.

- Be sure to verify that the instrument- rated voltage matches the supply voltage before power- on; otherwise, the power supply may be damaged.
- Always plug the main power cord into a power outlet with protective earth grounding. Do not use extension sockets without protective earth. Ensure the instrument is properly grounded before operation.
- Observe the positive and negative polarity markings before wiring the instrument; incorrect connection will damage the instrument.

Self-check Steps

The normal self-check procedure for the instrument is as follows:

1. Connect the power cord correctly, press the power key to power on, and the instrument will perform self- test.
2. After completing the instrument self-test, the TFT display shows the following status.



Exception handling

The instrument does not start properly when activated. Follow the steps below to check and resolve the issue.

1. Check whether the power cord is connected correctly and confirm that the instrument is powered on.

Power cable connection is good => 2

Power connection error => Reconnect the power cord and check if the issue resolves.

2. Is the instrument powered on? Pressing the power button sets the device to the powered -on state.

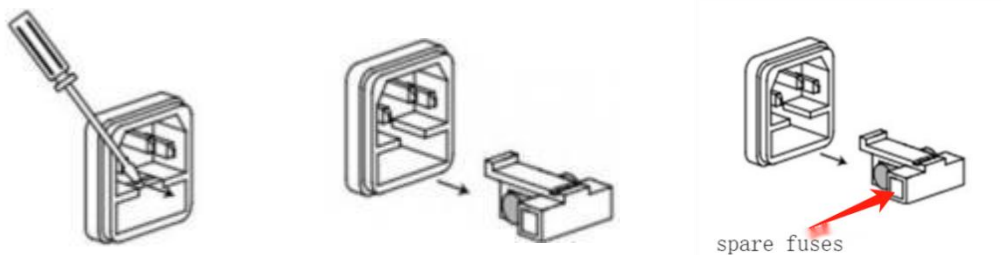
Yes => 3

If not: Please turn on the device by pressing the power button and check if the anomaly has been resolved.

3. Check whether the fuse of the instrument is damaged.

If the fuse is burned out, replace it. Follow these steps:

- 1) Unplug the power cord and remove the fuse box using a small screwdriver (the fuse location is described on the back panel).



2) Replace with a fuse of the same specification, place it in the box, and reinstall.

Fuse Specification Details Table:

Model	Fuse Specifications (220V)	Fuse Specifications (110V)
MPS-31,MPS-32	3.15A	5A
MPS -33,MPS-36, MPS-38,MPS-39	5A	6.3A
MPS-34, MPS-35, MPS-37	8A	10A

2.6 Output Check

The output check ensures that the power supply achieves its rated output and responds correctly to front- panel operations.

Output Voltage Check

Verify the basic voltage performance of the power supply under no- load condition.

1. Turn on the power supply.
2. Set the power supply current value (≥ 0.1 A).
3. Enable power supply output. Press the Output key; the ON and CV status indicators on the TFT display will light up.
4. Set the power- supply voltage. Set different power- supply voltages, and check whether the voltage value shown on the TFT is close to the set value and whether the displayed current value is close to 0 A.
5. Confirm that the power- supply voltage can be adjusted from 0V up to the maximum output voltage.

- - - End

Output Current Check

Verify the basic current performance of the power supply under output short- circuit condition.

1. Turn on the power supply.
2. Disable power supply output and confirm it is in OFF state with the OFF status indicator shown on the TFT.
3. Connect an insulated wire between the positive (+) and negative (–) output terminals of the power supply.

(The wire used must be capable of carrying the maximum output current of the power supply.)

4. Set the power- supply voltage to 1 V.
 5. Enable the power- supply output.
 6. Set the power- supply current. Set different power- supply current values and check whether the current reading on the TFT is close to the set value.
 7. Verify that the power- supply current can be adjusted from 0 A up to the rated full- output current of this channel.
 8. Disable the power- supply output and remove the short- circuit wire.
- - - End

Chapter 3: Functions and Features

This chapter describes the functions and characteristics of the power supply in detail. It consists of the following sections:

- Switching between Local / Remote Operation
- Voltage Setting Operation
- Current Setting Operation
- Output On- Off Setting Operation
- Over- Voltage and Over- Current Protection Functions
- Store- and- Recall Operation Function
- List Operation
- Menu Function
- Keypad Lock Function

3.1 Switching between Local / Remote Operation

The power supply provides two operation modes: Local and Remote. Switching between the two modes can be performed via communication commands. The power supply defaults to Local mode upon initialization.

Local Operation Mode: Perform relevant operations using keys on the power- supply front panel.

Remote Operation Mode: The power supply is connected to a PC, and relevant operations are executed on the PC through communication commands. When the power supply is in Remote mode, all panel keys except the Local key are disabled. Press the Local key to switch back to Local mode. Changing the operation mode will not affect the output parameters of the power supply.

3.2 Voltage Setting Operation

3.2.1 The CH1 voltage setting range is from 0 V to the maximum output voltage. When the cursor is at the voltage- setting position, move the cursor with the left/right keys or by pressing the voltage knob; rotate the voltage knob to adjust the set- voltage value. When the cursor is not at the voltage- setting position, use the up/down direction keys to move the cursor.

3.2.2 CH2 has four fixed voltage values: 1.8 V, 2.5 V, 3.3 V and 5 V. These values are non- adjustable and can only be switched among the four groups.

CH2 voltage can only be set when the “CH2” cursor on the TFT display is highlighted. Use the up/down direction keys to move the cursor to the “CH2” position, then rotate the voltage knob to switch voltage values. The currently selected voltage value is shown below CH2.

3.3 Current Setting Operation

3.3.1 The CH1 current setting range is from 0 A to the maximum output current value.

When the cursor is at the current- setting position, press the current knob to move the cursor, and rotate the current knob to adjust the set current value. When the cursor is not at the current- setting position, use the up- down keys to move the cursor.

3.3.2 CH2 current is fixed at 3 A and non- adjustable.

3.4 Output On-Off Setting Operation

Press the Output key on the front panel to turn the CH1 and CH2 outputs of the power supply on or off. When the output is enabled, the operating- status indicators (ON and CV/CC) on the TFT display light up, and the timer in the upper- right corner resets and starts counting. When the output is disabled, the OFF status indicator on the TFT lights up, and the timer in the upper- right corner stops counting.

Note

Connect the power supply to the device- under- test before pressing the Output key to enable output; otherwise arcing may occur during connection. If no output is available after output is turned on, check the voltage and current settings. Set both voltage and current to non- zero values before enabling output again.

3.5 Over-Voltage / Over-Current Protection Function

The OVP/OCP function allows users to set protection thresholds. When the voltage or current of the device-under-test exceeds the set threshold, the power supply activates OVP/OCP protection. Upon OVP or OCP trigger, the power- supply output shuts down, and relevant alarm prompts (OVP/OCP) are shown on the TFT display.

3.5.1 OVP (Over-Voltage Protection)

Use the up/down keys to move the cursor to the over- voltage setting item. The OVP status (OFF or ON) is highlighted. Rotate the voltage knob left or right to enable or disable the protection. Press the down key to highlight the OVP set-value parameter. Move the cursor with left/right keys or by pressing the voltage knob, then rotate the voltage knob to configure the OVP threshold.

3.5.2 OCP (Over-Current Protection)

Use the up/down keys to move the cursor to the over-current setting item. The OCP status (OFF or ON) is highlighted. Rotate the current knob left or right to enable or disable the protection. Press the down key to highlight the OCP set- value parameter. Move the cursor with left/right keys or by pressing the current knob, then rotate the current knob to configure the OCP threshold.

3.6 Store- and- Recall Operation Function

There are four memory keys on the keypad: M1, M2, M3 and M4.

The power supply can store frequently- used parameters into four non- volatile memory locations M1, M2, M3 and M4 for quick and convenient recall. After configuring parameters such as voltage and current, long- press any of the M1, M2, M3, M4 keys to save parameters to the corresponding memory slot. Saved parameters include voltage, current, OVP/OCP and other settings. The stored voltage and current values for each key are displayed at the bottom of the TFT screen. For recall, short- press the corresponding M1, M2, M3 or M4 key to load the saved parameters.

3.7 List Test Function

The List function allows you to edit a list file to set the value and duration for each step. After pressing the Output key, the power supply will automatically execute the configured value of each step.



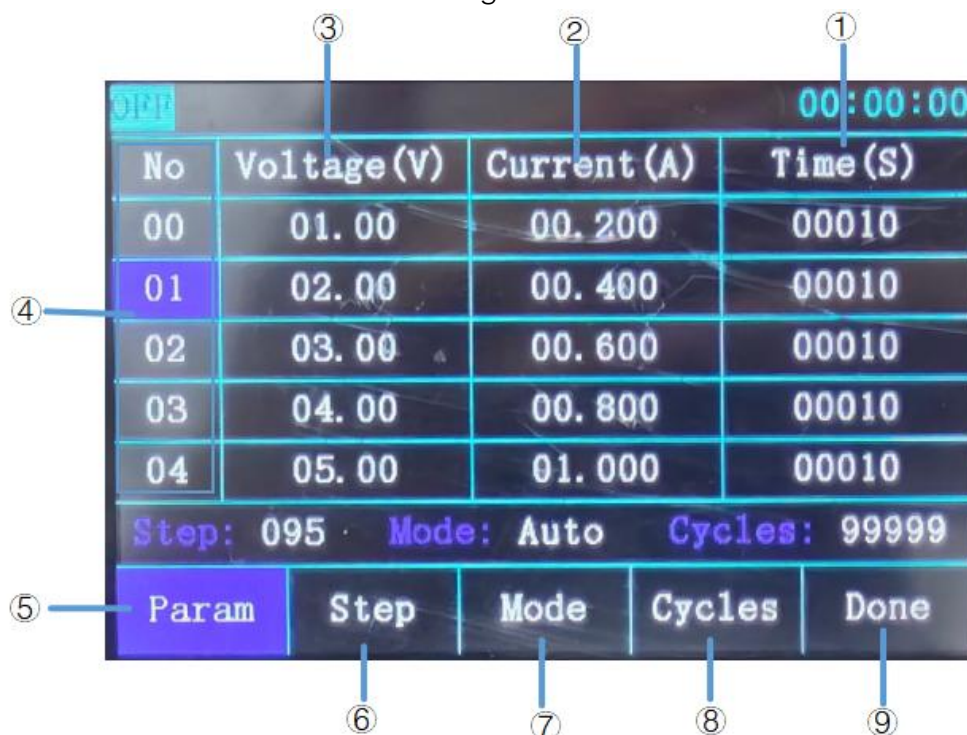
A. List sequence function switch: Select it with the left and right directional keys, then press the voltage knob to enable or disable the List function. After enabling the List function, press the List key once to return to the List main interface, as shown in the figure below:

- ①. Display of voltage and current setpoints for the current List step
- ②. Countdown for current- step execution

- ③. Display of voltage and current setpoints for the next List step
- ④. OVP switch and over- voltage value setting. Use the Up/Down keys to move the cursor to the OVP setting. Rotate the voltage knob to enable or disable over- voltage protection. Then use the Up/Down keys to move the cursor to the over- voltage parameter. Move the cursor with Left/Right keys, and rotate the voltage knob to modify the OVP value.
- ⑤. OCP switch and over- current value setting. Use the Up/Down keys to move the cursor to the OCP setting. Rotate the current knob to enable or disable over- current protection. Then use the Up/Down keys to move the cursor to the over- current parameter. Move the cursor with Left/Right keys, and rotate the current knob to modify the OCP value.
- ⑥. Display window for List output execution

B. List sequence edit setting: Select it with the left and right directional keys, then press the voltage knob to enter List sequence- editing operation. When the sequence function is enabled, the sequence edit settings cannot be modified; you must disable the sequence function first.

3.7.1 List Function Parameter Setting



No	Voltage (V)	Current (A)	Time (S)
00	01.00	00.200	00010
01	02.00	00.400	00010
02	03.00	00.600	00010
03	04.00	00.800	00010
04	05.00	01.000	00010

Step: 095 Mode: Auto Cycles: 99999

Param	Step	Mode	Cycles	Done
-------	------	------	--------	------

- ① Time: Dwell time of the current List step (0- 99999 seconds).
- ② Current: Maximum output current for the current List step.
- ③ Voltage: Output voltage for the current List step.
- ④ Step: Current step of List operation (0- 99 steps).
- ⑤ Parameter: This option is used to set parameters such as time, current and voltage for List operation. Press the voltage knob to select parameter setting. After entering the parameter setting page, rotate the voltage knob left or right to select the step to be edited. Press the voltage knob again to select editable parameters including time, voltage and current. Rotate the voltage knob left or

right to modify these values. Long- press the knob to shift the cursor digit position.

- ⑥ Total Steps: Total execution steps of the current List operation (0- 99 steps). This option sets the total number of List steps. Select it with left/right keys and modify by rotating the voltage knob.
- ⑦ Mode: This option sets the trigger mode for List operation. Only Auto mode is available currently. After pressing the Output key, each step will run automatically and sequentially with no additional operation required.
- ⑧ Cycle: This option sets the cycle count for List operation (0- 99999; 0 means infinite loop). Select it with left/right directional keys and modify by rotating the voltage knob. Short- press the voltage knob to shift digit position.
- ⑨ Done: After completing List parameter configuration, select "Done" using left/right keys and press the voltage knob to save settings and return to the List main interface.

3.8 Menu Function

Press the Menu key when the power supply is in standby state to enter the "Menu" menu function for setting relevant parameters of the power supply. After entering the "Menu" menu function, rotate the voltage knob left and right to move the cursor to the parameter to be edited, press the knob to enter parameter modification, rotate the knob to modify the selected parameter, and press the knob to confirm the modified parameter. After the parameter modification is completed, press the "Menu" key again to exit the menu function.



System Parameters

Power on: Setting for the power supply output state upon power- up. "On" means the power supply stays in output state when powered- on; "Off" means the power supply stays in off state when powered- on; "Last" means the power supply restores the output state before last power- off upon power- up.

PowerParm: Setting for parameter retention upon power- up. "Last" retains the data from last power- off; "Default" restores factory default values.

Beeper: Power supply keypad sound setting. ON enables the buzzer; OFF disables the buzzer.

Language: Power supply language setting. Chinese and English are available for selection.

FirstMode: Priority mode setting when power output is enabled. "CV" means voltage is activated first upon output enable; "CC" means current is activated first upon output enable.

Zoom in: Zoom setting for the power supply TFT display. After enabling this function, exit the menu interface. When no panel operation is performed for about 10 seconds, the TFT display switches to zoom- in mode, showing only voltage, current and power parameters. Press any key to temporarily exit the zoom- in screen and return to the main interface. The TFT display will re- enter zoom- in display when no panel operation occurs. Disable this function to turn off zoom display.

Configuration Parameters

Serial Port: Communication port setting.

1. Address: Power supply communication address setting, ranging from 1- 254.
2. Baud Rate: Available baud rates of the power supply: 4800, 9600, 14400, 19200, 38400, 56000, 57600, 115200.

Command:

1. Version: Power supply communication protocol type setting. Two protocols are available: SCPI and MODBUS.
2. EndChar: Power supply communication terminator setting. Available terminators: LF, CR, CRLF, LFCR.

Reset: Power supply reset. The reset setting provides "Yes" and "No" options. "Yes" resets the power supply; "No" skips reset. This reset initializes the power supply Menu settings to factory defaults. The settings after initialization are as follows:



3.9 Keypad Lock Function

Press the Lock key to lock the power supply keypad. All keys and the knob become invalid except the Lock key. Long- press again to unlock the keypad.

Chapter 4: Technical Specifications

Model		MPS-31	MPS-32	MPS-33	MPS-34	MPS-35	MPS-36	MPS-37	MPS-38	MPS-39
Rated output voltage		0~30V	0~30V	0~30V	0~30V	0~30V	0~60V	0~60V	0~80V	0~100V
Rated output current		0~2A	0~4A	0~6A	0~8A	0~10A	0~2A	0~4A	0~2A	0~0.5A
Rated output power		60W	120W	180W	240W	300W	120W	240W	160W	50W
Load regulation	Voltage	0.05%+5mV	0.05%+5mV	0.05%+5mV	0.05%+8mV	0.05%+8mV	0.05%+5mV	0.05%+5mV	0.05%+5mV	0.05%+5mV
	Current	0.1%+5mA	0.1%+5mA	0.1%+5mA	0.1%+8mA	0.1%+5mA	0.1%+5mA	0.1%+5mA	0.1%+5mA	0.1%+5mA
Line regulation rate	Voltage	0.05%+8mV	0.05%+5mV	0.05%+8mV	0.05%+10mV	0.05%+5mV	0.05%+8mV	0.05%+5mV	0.05%+5mV	0.05%+5mV
	Current	0.1%+5mA	0.1%+5mA	0.1%+5mA	0.1%+8mA	0.1%+5mA	0.1%+5mA	0.1%+5mA	0.1%+5mA	0.1%+5mA
Setting resolution	Voltage	10mV	10mV	10mV	10mV	10mV	10mV	10mV	10mV	10mV
	Current	1mA	1mA	1mA	1mA	1mA	1mA	1mA	1mA	1mA
Read back resolution	Voltage	10mV	10mV	10mV	10mV	10mV	10mV	10mV	10mV	10mV
	Current	1mA	1mA	1mA	1mA	1mA	1mA	1mA	1mA	1mA
Setting accuracy (25 °C±5°C)	Voltage	≤0.1%+2digits	≤0.1%+2digits	≤0.1%+2digits	≤0.1%+2digits	≤0.1%+2digits	≤0.1%+2digits	≤0.1%+3digits	≤0.1%+3digits	≤0.1%+3digits
	Current	≤0.2%+3digits	≤0.2%+3digits	≤0.2%+3digits	≤0.2%+6digits	≤0.2%+3digits	≤0.2%+3digits	≤0.2%+3digits	≤0.2%+3digits	≤0.2%+3digits
Read back accuracy (25 °C±5°C)	Voltage	≤0.1%+2digits	≤0.1%+2digits	≤0.1%+2digits	≤0.1%+2digits	≤0.1%+2digits	≤0.1%+2digits	≤0.1%+3digits	≤0.1%+3digits	≤0.1%+3digits
	Current	≤0.2%+3digits	≤0.2%+3digits	≤0.2%+3digits	≤0.2%+6digits	≤0.2%+3digits	≤0.2%+3digits	≤0.2%+3digits	≤0.2%+3digits	≤0.2%+3digits
Ripple and noise (25°C±5 °C)	Voltage	3mVrms	3mVrms	5mVrms	5mVrms	5mVrms	5mVrms	5mVrms	5mVrms	10mVrms
	Current	3mA _{rms}	3mA _{rms}	3mA _{rms}	6mA _{rms}	3mA _{rms}	3mA _{rms}	3mA _{rms}	3mA _{rms}	3mA _{rms}

4.1 Main Technical Specifications

4.2 Additional Features

Recommended calibration frequency: once per year

Cooling method: Forced air cooling

Operating environment temperature: 0 to 40 °C

Storage temperature: -20 to 70 °C

Usage Environment: Designed for indoor use, pollution level 2, maximum humidity 80%

Chapter 5: Communication Between Power Supply and PC

Communication between Power Supply and PC

The power supply supports optional communication interfaces: RS232, USB, RS485 and RJ45 (Ethernet/LAN). Users may select one of them for communication with a computer.

Before communication operations, you shall first match the following parameters between the instrument and the PC.

1. Select the required communication interface type (available as RS232 or LAN port).
2. Baud Rate: 9600 (available values: 4800, 9600, 14400, 19200, 38400, 56000, 57600, 115200). You may enter the system menu via the front- panel to set the communication baud rate.
3. Protocol: Set the instrument communication protocol type; two options are available: SCPI or MODBUS.
4. Terminator: Setting for instrument communication terminator. Available terminators: LF, CR, CRLF, LFCR. (The terminator must be correct for SCPI protocol.)
5. Address: Set the instrument communication address, ranging from 1 to 254 (the address must be correct when using the MODBUS protocol).

Data bits: 8

Stop position: 1

parity: NONE (all 8 data bits have no parity)

Start Bit	8 Data Bits	Parity=None	Stop Bit
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